

Region Affinities

Methods Primer

A Guide to the Statistical and Network Methods

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*Companion documents: Summary · Technical Appendix · Data Appendix · Region Reference —
Identity · Region Reference — Terroir*

Introduction

The Region Affinities tool uses a sequence of methods from natural language processing, multivariate statistics, and network analysis to convert two parallel descriptions of 59 wine regions — their cultural identity narratives and their structured terroir profiles — into two independent classification systems, and to test whether those classifications agree. This primer explains each method in the order it appears in the analytical pipeline. The treatment assumes comfort with elementary statistics (means, standard deviations, hypothesis testing) but no prior exposure to text vectorisation, dimensionality reduction, or cluster analysis.

Each section explains the intuition behind a method, gives its standard form, and notes what it reveals in the specific context of Region Affinities. Full mathematical treatment is in the companion *Technical Appendix*.

1. TF-IDF: Turning Text into Numbers

What it does. TF-IDF (term frequency–inverse document frequency) converts each region’s terroir profile from prose into a numerical vector that downstream algorithms can analyse. The intuition is straightforward: a word that appears frequently in one region’s profile but rarely across the others is probably distinctive of that region; a word that appears in many profiles (*wine, region, vineyard*) is not.

The two halves. The *TF* half counts how often each term appears in each profile; the *IDF* half down-weights terms that appear in many profiles. The product of the two gives each (region, term) pair a score: high for distinctive terms, low for common ones. The full collection of scores forms a matrix in which each row is a region and each column is a term; this matrix is the input to the next stage.

In Region Affinities. The TF-IDF vocabulary spans both single words and two-word phrases (e.g., *slate* and *slate slope, volcanic* and *volcanic ash*), capped at 800 features to keep the matrix manageable. The matrix is sparse: most regions use only a small fraction of the vocabulary. This sparsity is what motivates the next step.

2. Principal Component Analysis (PCA)

What it does. PCA compresses high-dimensional data into a smaller number of dimensions while preserving as much of the meaningful variation as possible. After TF-IDF, each region is described by 800 numbers; after PCA, each region is described by 10. The 10 PCA dimensions are not chosen by hand — they are constructed by the algorithm to be the directions of greatest variance in the original 800-dimensional space.

The intuition. Many TF-IDF terms vary together. *Granite*, *schist*, and *steep* tend to co-occur in profiles of mountain wine regions; PCA combines them into a single dimension that captures that pattern. *Riesling*, *slate*, and *steep slope* similarly co-vary in the German Rhine profiles; PCA combines those too. Each principal component is a linear combination of original terms that captures one coherent pattern of variation.

Variance retained. The first 10 components capture 32.2% of the total variance in the terroir matrix. This may sound low, but it is typical for sparse, high-dimensional text representations: most of the apparent “variance” in the original 800-dimensional space comes from terms that appear in only one or two profiles and contribute noise rather than signal. The 10-component projection retains the signal and discards the noise.

For the identity layer. PCA is also applied to the identity layer, but here the original space is already only six-dimensional (one per identity dimension D1–D6), so all six components are retained and the projection captures 100% of the variance. The role of PCA in the identity layer is therefore purely a rotation into the natural axes of variation, not a compression.

3. K-Means Clustering

What it does. K-means partitions a collection of points into a specified number of groups (clusters) such that each point is in the cluster whose centre is nearest. The algorithm is iterative: it begins with K randomly chosen centres, assigns each point to its nearest centre, recomputes each centre as the mean of its assigned points, and repeats until the assignments stop changing.

Choosing K . The number of clusters K is a modelling choice. The standard guide is the silhouette score (Section 4): one chooses the K that produces the cleanest cluster separation. For Region Affinities, the identity layer uses $K = 6$ and the terroir layer uses $K = 7$. Both choices were made by silhouette maximisation across $K \in \{3, \dots, 10\}$.

Random initialisation. Because K-means is sensitive to its starting centres, the algorithm is run 20 times with different initialisations and the result with the lowest within-cluster sum of squares is retained. The random seed is fixed for reproducibility.

In Region Affinities. The terroir clustering is fully model-derived: no human input is involved in deciding which terroir cluster a region belongs to, only in pre-deciding K . The identity clustering is also fully algorithmic given the D-scores; the only human input is the assignment of D-scores to each region from the corresponding identity narrative.

4. Silhouette Score

What it does. The silhouette score measures how well-defined the resulting clusters are. For each region, it compares the average distance to other regions in the same cluster (cohesion) with the average distance to regions in the nearest other cluster (separation). The result is a number between -1 and $+1$:

- $+1$: the region is firmly inside its own cluster, far from any other cluster.
- 0 : the region sits on the boundary between two clusters.
- -1 : the region is closer to a different cluster than to its own (suggesting misclassification).

The overall silhouette score is the mean across all regions. In social-science data, scores above 0.20 indicate meaningful structure; scores above 0.30 indicate clear separation.

For Region Affinities. The identity layer has silhouette 0.303 (clear structure); the terroir layer has silhouette 0.246 (meaningful structure, but tighter clusters in some regions and looser in others). Both are within the expected range for descriptive cultural data.

5. Cosine Similarity and Nearest-Neighbour Graphs

What it does. For each layer, every pair of regions is assigned a similarity score between 0 and 1, computed as the cosine of the angle between their feature vectors. Two regions whose profiles point in the same direction in feature space have cosine similarity close to 1; two regions with unrelated profiles have cosine similarity close to 0.

Why cosine. Cosine similarity ignores the magnitude of the vectors and looks only at their direction. For TF-IDF vectors this is critical: a longer profile will have larger raw counts than a shorter one, but cosine similarity correctly identifies them as similar if the same words dominate both. For D-score vectors it is also natural: a region with all dimensions at $+2$ is structurally similar to one with all dimensions at $+1$, even though their raw magnitudes differ.

The k -nearest-neighbour graph. For visualisation, only the strongest similarities are kept. Each region is connected by an edge to its five most similar peers in the layer; the result is a sparse graph that is much easier to read than the full 59×59 similarity matrix. The graph is symmetrised: an edge between two regions exists if either of them is among the other's top five.

In Region Affinities. Two such graphs are constructed — one for the identity layer, one for the terroir layer — and shown side by side in the Dual Networks view. Comparing the neighbour sets across the two graphs is the central interactive activity of the tool.

6. Adjusted Rand Index (ARI)

What it does. The Adjusted Rand Index measures how much two clusterings of the same set of objects agree. It looks at every pair of objects and asks: do the two clusterings agree

on whether this pair belongs together or apart? After counting agreements across all pairs, it adjusts for the agreement that would be expected by random chance.

The scale.

- $\text{ARI} = 1.0$: the two clusterings are identical.
- $\text{ARI} = 0.0$: the two clusterings agree no more than two random labellings of the same objects would be expected to agree.
- $\text{ARI} > 0.5$: substantial agreement, common in well-aligned classification systems.
- $\text{ARI} < 0.1$: the two clusterings are essentially independent.

For Region Affinities. ARI between the identity clustering ($K = 6$) and the terroir clustering ($K = 7$) is 0.023. This is statistically indistinguishable from zero. The two classifications behave as if they were drawn independently of each other, even though both are derived from the same 59 regions.

7. Chi-Squared Test of Independence

What it does. The chi-squared test is a complementary check on the same independence question. It builds a 6×7 contingency table whose rows are identity clusters, columns are terroir clusters, and cells count how many regions fall into each combination. Under the null hypothesis that identity and terroir cluster labels are statistically independent, each cell’s expected count is the product of the row and column marginal totals divided by the grand total. The test compares observed counts to expected counts.

Reading the result. The chi-squared test produces a p -value: the probability of observing a deviation from independence as large as the one observed, if the two labellings were truly independent. A small p -value (conventionally below 0.05) is grounds to reject independence; a larger p -value is consistent with independence.

For Region Affinities. The chi-squared statistic is 38.78 on 30 degrees of freedom; the p -value is 0.131. The test fails to reject independence at the conventional 5% level, in agreement with the ARI result.

8. The Pipeline in Summary

The full Region Affinities pipeline can be read as a sequence of five stages:

1. *Encoding.* The identity layer is encoded as 6-dimensional D-score vectors (one per region, scored from the identity narrative). The terroir layer is encoded as TF-IDF vectors (one per region, computed from the structured terroir profile).
2. *Reduction.* Each layer is standardised and projected via PCA into a low-dimensional dense representation suitable for clustering (6 components for identity, 10 for terroir).

3. *Clustering.* K-means partitions each layer into $K_{\text{id}} = 6$ identity clusters and $K_{\text{terr}} = 7$ terroir clusters. The silhouette score validates each partition’s quality.
4. *Visualisation.* For each layer, a k -NN similarity graph with $k = 5$ is built from cosine similarities and rendered as an interactive force-directed network.
5. *Independence test.* The ARI and the chi-squared test on the 6×7 contingency table evaluate whether the identity and terroir cluster labels carry information about each other. They do not.

The empirical conclusion — that identity and terroir are statistically independent classification systems for wine regions — emerges from the agreement between the two independence measures, the stability of the result across reasonable changes to K , and the visible patterns of disagreement at the level of individual regions in the Dual Networks view.

Further Reading

For PCA: Jolliffe, I. T. (2002). *Principal Component Analysis* (2nd ed.). Springer. Chapters 1–3 are accessible to readers comfortable with basic linear algebra.

For K-means and clustering more broadly: Hastie, T., Tibshirani, R. & Friedman, J. (2009). *The Elements of Statistical Learning* (2nd ed.). Springer. Chapter 14.

For ARI and cluster comparison: Hubert, L. & Arabie, P. (1985). Comparing partitions. *Journal of Classification*, 2(1), 193–218.

For TF-IDF and text representation: Salton, G. & Buckley, C. (1988). Term-weighting approaches in automatic text retrieval. *Information Processing & Management*, 24(5), 513–523.

For network analysis: Newman, M. E. J. (2010). *Networks: An Introduction*. Oxford University Press.